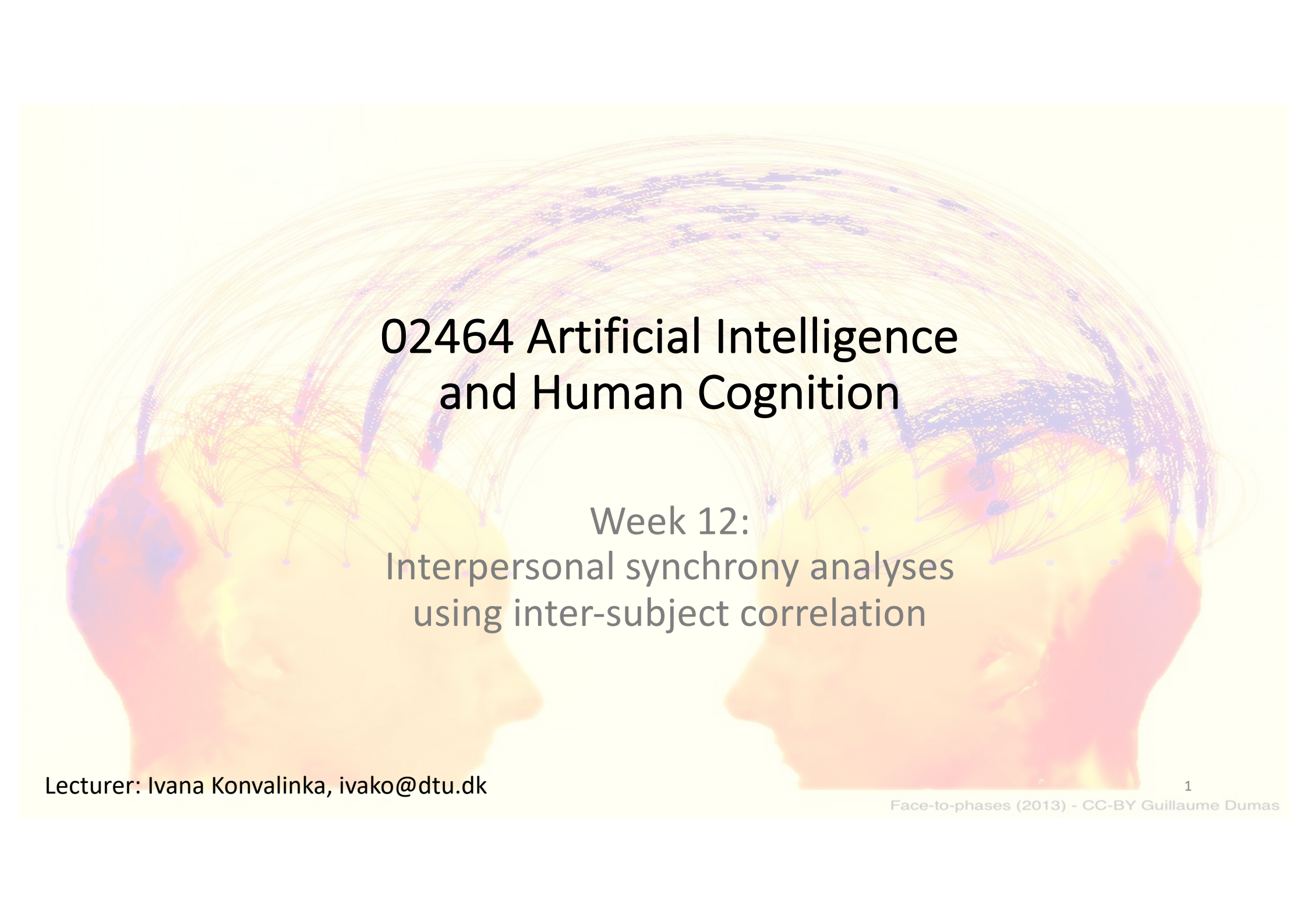




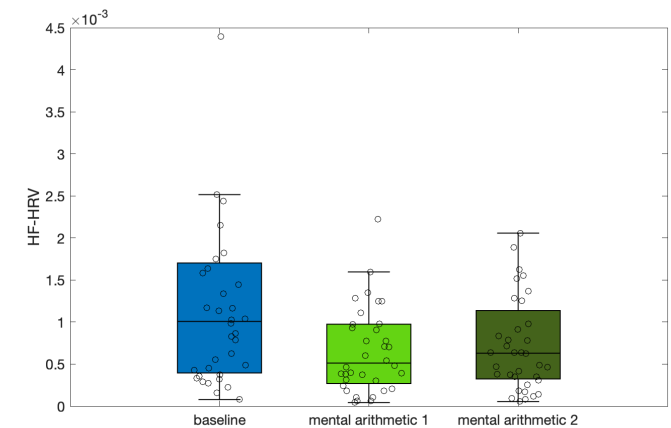
02464 Artificial Intelligence and Human Cognition

Week 12:
Interpersonal synchrony analyses
using inter-subject correlation

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Mini-project 3 tasks

- Part I: HRV analysis in the frequency domain
 - Estimate the power spectrum of HRV using Welch's method (time -> frequency domain)
 - Estimate PSD by integrating across high frequencies (0.15-0.4 Hz)
 - Run the analysis across all the participants (you can pick one group: session A or B!)
 - Plot the PSD of HF-HRV across each of your conditions (baseline, game 1, game 2, game 3), using e.g., box-plots, or violin plots, or raincloud plots
 - Do 3 Wilcoxon signed-rank tests to see whether HF-HRV is different between baseline vs. game1, baseline vs. game2, baseline vs. game 3
 - Interpret the results



Last year's results:

- HF-HRV was lower during mental arithmetic compared to baseline
- No differences between easy and difficult mental arithmetic conditions

Part II: Inter-subject correlation

Pearson's correlation coefficient

- Pearson's r coefficient is a statistical method that measures the strength of the linear association between two variables, ranging from -1 to $+1$
- The closer r is to $+1$ or -1 means stronger association (+ means that as one variable increases, so does the value of the other variable; $-$ means as one variable increases, the other decreases), 0 means no association

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where,

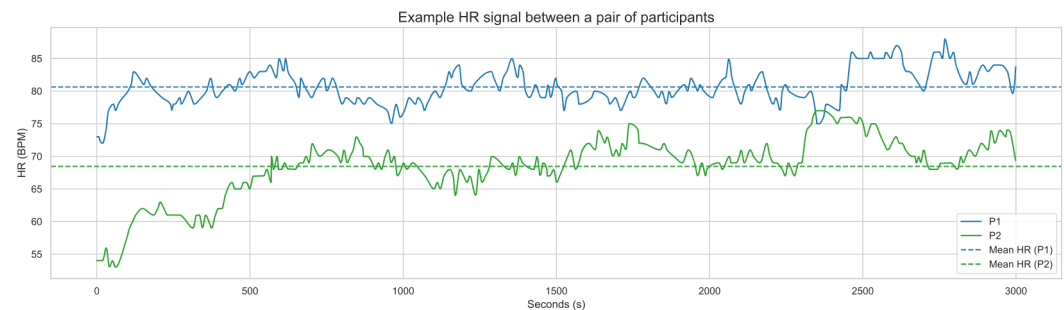
r = Pearson Correlation Coefficient

x_i = x variable samples

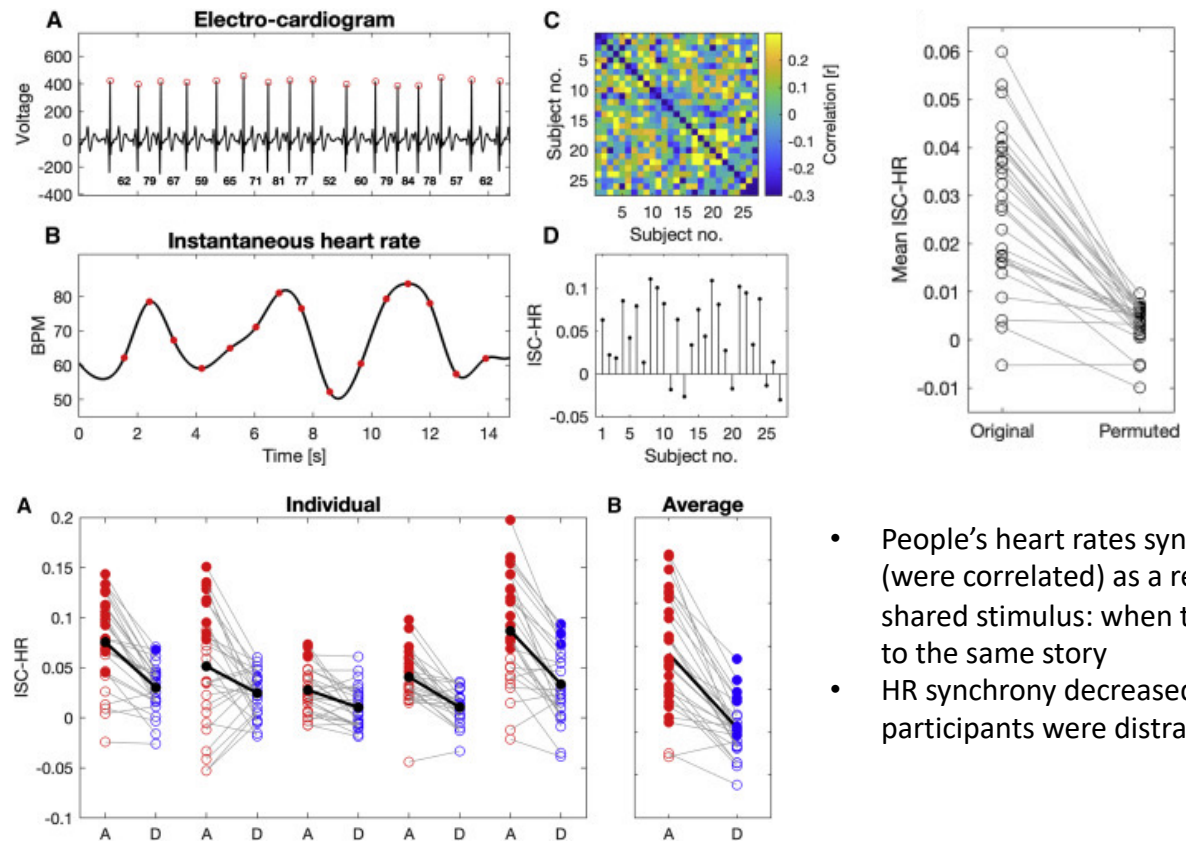
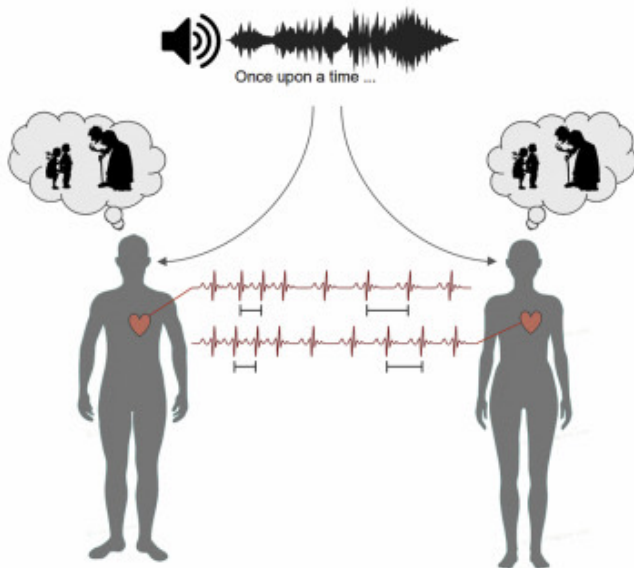
y_i = y variable sample

$\bar{x}$ = mean of values in x variable

$\bar{y}$ = mean of values in y variable

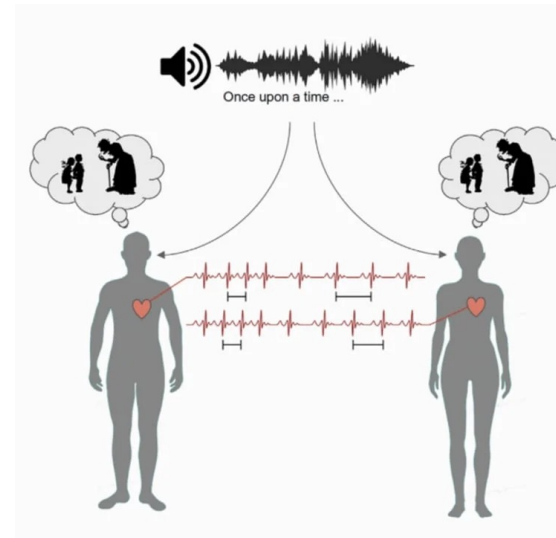


Inter-subject correlation: heart rate synchrony when listening to the same narratives



- People's heart rates synchronized (were correlated) as a result of shared stimulus: when they listened to the same story
- HR synchrony decreased when participants were distracted

Heart rate synchrony reflects joint attention and engagement



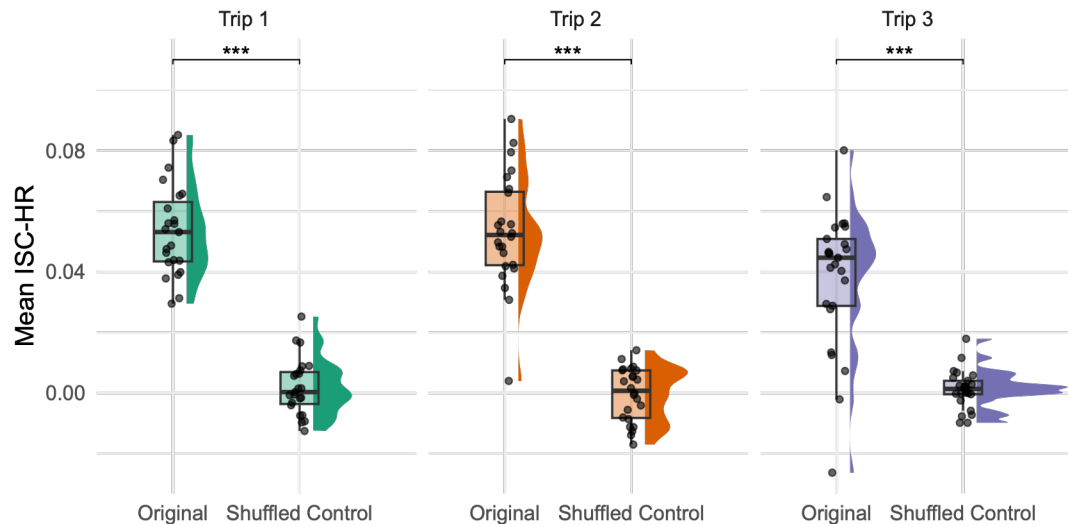
Watching the same movie or listening to the same music or story aligns heart rhythms. This synchrony happens even without interaction or communication. It is driven by shared sensory input.

Synchrony is greater when we are together, and also when we simultaneously focus our attention. Hence, HR synchrony is considered a marker joint attention and engagement.

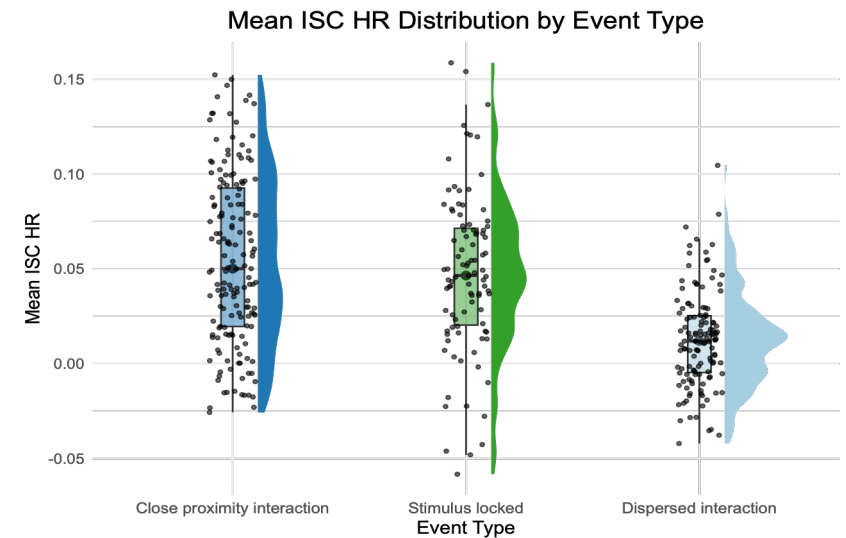
Heart-rate synchrony is modulated by social and environmental factors in real-world settings



People synchronize heart rates when in close proximity

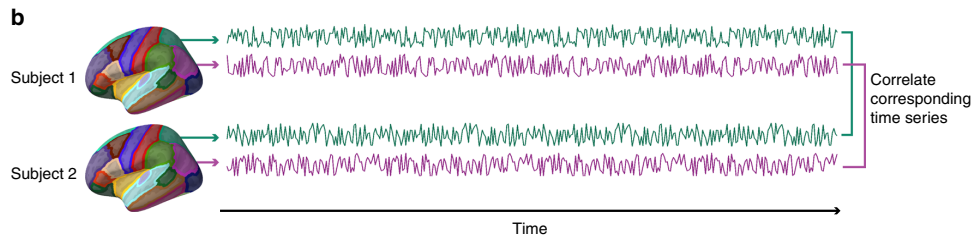
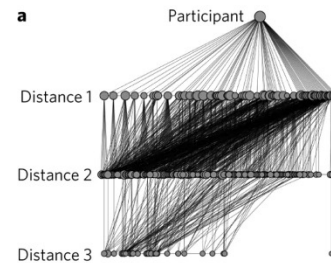


Synchrony depends on the social context

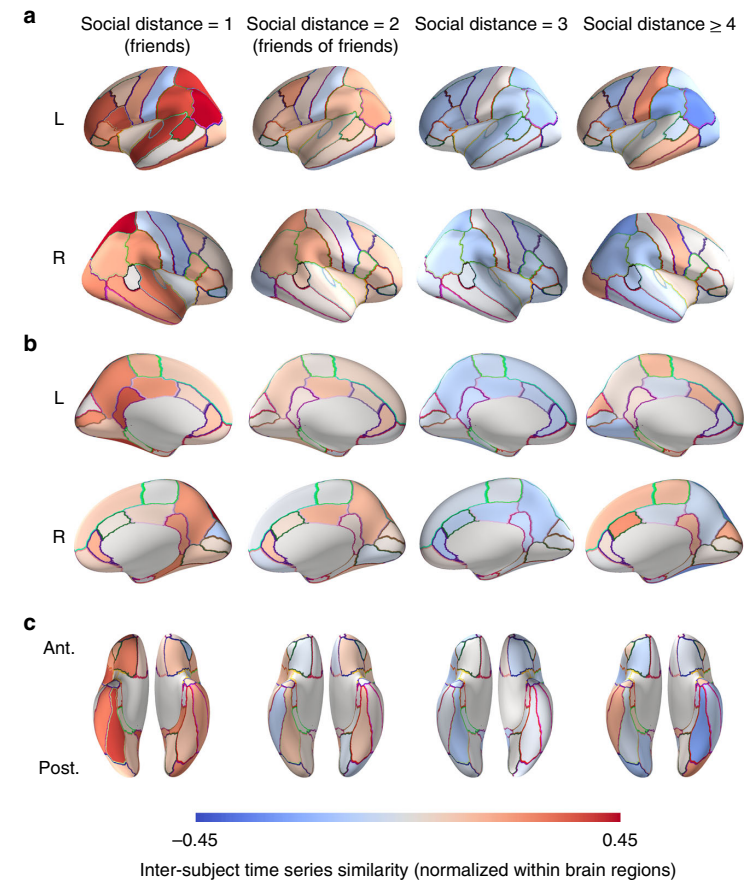


He, Sørensen, Christensen, Konvalinka (in prep)

Inter-subject correlation between neural signals is modulated by friendship



- Friends had more correlated/similar neural responses when watching the same movies, and inter-subject correlation on a neural level decreased with increasing social distance



Interpersonal physiological and neural synchrony

- Studies show that people synchronize their physiological rhythms when they engage in interaction (action-perception coupling), but also when locked to the same stimulus – which is modulated by joint attention and engagement
- Friends have more correlated neural rhythms when passively watching the same movies – does this reflect that they are engaged in more similar ways?
- Do friends synchronize bodily rhythms more with each other than with non-friends during game-based social interactions?
 - What underlying mechanisms might drive this result?

Mini-project 3 tasks

- Part II: interpersonal heart-rate synchrony
 - Compute inter-subject correlations between RR-time series of team members and non-team members, i.e., non-interacting surrogate pairs (calculating Pearson r 's between pairs of time-series) for games 2 & 3
 - Plot inter-subject correlation distributions for team members and non-team members for the two games (using box-plots, or violin plots, or raincloud plots)
 - Test whether there is more heart rate synchrony between team members compared to non-team members, use 2 Mann-Whitney U tests to compare the two groups for games 2 & 3
 - Now do the same procedure as above to compare inter-subject correlations between friends and non-friends (ONLY FOR SESSION A)
 - Interpret both sets of results